

Shared Task on Automatic Evaluation of Scientific Writing

Dan Flickinger, Michael Goodman, and Woodley Packard

Stanford and UW

DELPH-IN Summit

CSLI, Stanford

17 June 2016

The Task

Predict whether a given sentence requires editing to ensure its 'fit' within the scientific writing genre.

Sample text:

```
<sentence sid="919.0">Thus there <del>exist</del>  
<ins>exists</ins> a constant _MATH_ such that  
_MATH_ for all _MATH_.</sentence>
```



Peering into the corpus

- First searched training corpus to find frequent correction types
- One of every three sentences contains at least one correction

Error Type	Count	% Lines
del-only	124,919	9.03%
ins-only	267,465	17.89%
del-ins	341,108	28.68%
total	733,492	39.24%



Most frequent error types

,	40.09%	<ins>,</ins>	52.94%	 <ins>-</ins>	13.03%
the 	9.19%	<ins>the </ins>	12.66%	-<ins> </ins>	5.10%
:	8.96%	<ins>.</ins>	4.78%	,<ins>;</ins>	1.92%
-	3.14%	<ins>a </ins>	4.03%	.<ins>,</ins>	0.74%
.	1.94%	<ins>:</ins>	2.87%	section<ins>Section</ins>	0.66%
's	1.84%	<ins>that </ins>	2.14%	:<ins>.</ins>	0.60%
that 	1.49%	<ins>and </ins>	1.92%	the<ins>a</ins>	0.59%
a 	1.45%	<ins>an </ins>	0.78%	a<ins>the</ins>	0.56%
'	1.29%	<ins>is </ins>	0.63%	which<ins>that</ins>	0.54%
"	0.91%	<ins>of </ins>	0.63%	is<ins>are</ins>	0.54%



Our method

- Use the ERG augmented with mal-rules
- Parse using ACE
- Use most likely derivation (just using Redwoods-trained model)
- If the root signals a 'robust' analysis, flag as needing editing



Sample derivation tree with mal-rule

```
(sb-hd_mc_c
(sp-hd_n_c
(another "another")
(hd-cmp_u_c
(aj-hdn_norm_c
(hd_optcmp_c
(j_tough_dlr (important_a2 "important")))
(n_sg_ilr (point_n2 "point"))))
(hd-cmp_u_c
(to_c_prop "to")
(hd_xcmp_c
(v_n3s-bse_ilr (note_v1 "note"))))))))
(hd-cmp_u_c
(be_id_is "is")
(sp-hd_n_c
(the_1 "the")
(hd-cmp_u_c
(n_sg_ilr (problem_n1 "problem"))
(hd-cmp_u_c
(of_prtcl "of")
(hdn.bnp.c.rbst
(aj-hdn_norm_c
(j_j-un_dlr
(v_j-nb-pas-tr_dlr
(v_pas_odlr (know_v1 "unknown")))))
(np-hdn_cpd_c
(hdn_bnp-pn_c
(n_sg_ilr (generic_proper_ne "SNR")))
(w_period_plr
(n_sg_ilr
(situation_n1 "situation."))))))))))
```



Sample mal-rules added to the ERG

a. Syntactic: Comma-spliced sentences

cl-cl_runon-cma_c_rbst

Multi-biometrics may address the problem of nonuniversality, e.g., in a speaker recognition system, the individuals who cannot speak cannot be enrolled.

b. Lexical: Subject-verb agreement mismatch

third_sg_fin_v_rbst

In what follow, this letter investigates mixed synchronization of fractional-order Lorenz-like system

c. Lexical entry: Missing obligatory direct object

allow_v1_rbst

The LHC will be a top quark factory, allowing to study several of its properties in great detail.



System results: Probabilistic track

Team	F-Score	Correlation
HITS	0.8311	0.0600
UW-SU	0.7849	0.2471
ISWD	0.7581	0.2690
NTNU-YZU	0.7419	0.4043
ISWD	0.7224	0.1298
HITS	0.7220	0.1666
NTNU-YZU	0.6926	0.4173
Knowlet	0.6925	0.3516



System results: Probabilistic track

Estimator	F-Score	Accuracy
B	0.5411	0.6333
output _{boolean}	0.5854	0.5322
no errors	—	0.6111
all errors	0.5600	0.3889



An adjustment for the F-score metric

Estimator	F-Score	Correlation
$E(B I)$	0.5644	0.2414
output _{probabilistic}	0.7902	0.2602
$P = 0.63$	0.7756	—



System results: Boolean track

Team	Prec	Rec	F-score
HU	0.5444	0.7413	0.6278
NTNU-YZU	0.5025	0.7785	0.6108
ISWD	0.4482	0.7279	0.5548
UW-SU	0.4145	0.8201	0.5507
UDE-LTL	0.3851	0.9241	0.5436
HITS	0.3765	0.9480	0.5389
ISWD	0.3960	0.6970	0.5051
NTNU-YZU	0.6717	0.3805	0.4858
Knowlet	0.6241	0.3685	0.4634



Some vexations in the data

- Lossy substitution of math and citation expressions
- Inconsistencies in normalizing use of commas and hyphens

“ second order ”	636
“ second<ins>-</ins>order ”	710
“ second-order ”	952

